

#3D251F
cocap

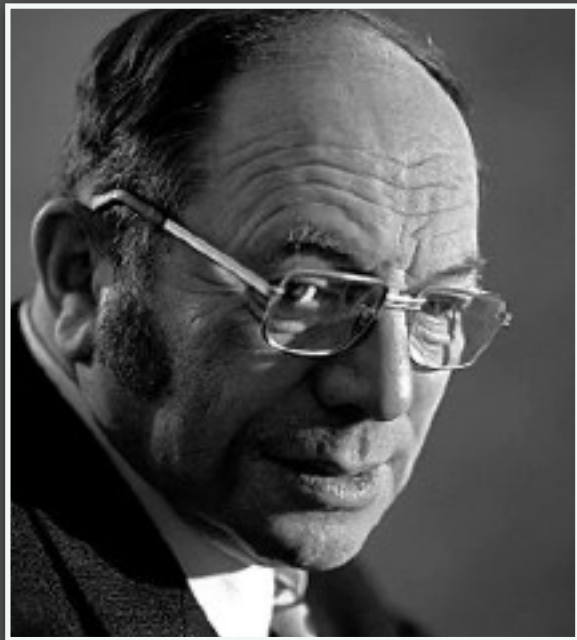
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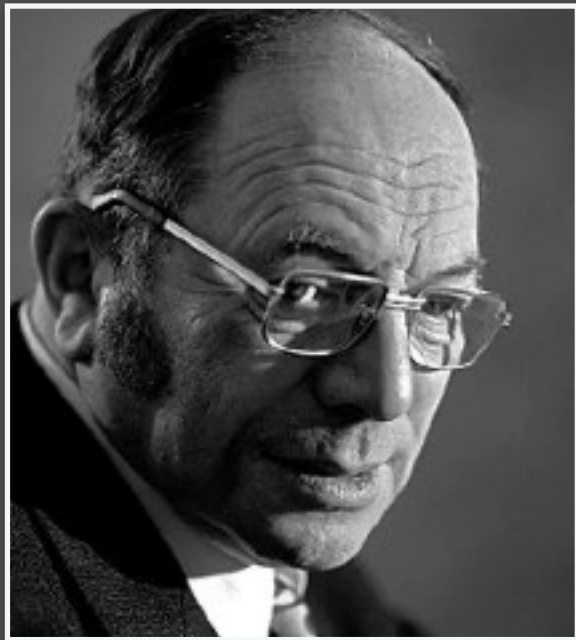


Kantorovich's Problem:

(Continuous version primal)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d\pi(x, y)$$

Key Takeaway: gives geometry on prob.
distributions, finite, meaningful, stable



Kantorovich's Problem:

(Continuous version dual)

$$\sup_{f, g : f \oplus g \leq c} \int_X f \, d\alpha + \int_Y g \, d\beta$$

Brenier's Theorem Corollary: The optimal dual potentials and φ are related by:

$$\varphi(x) = \frac{1}{2}\|x\|^2 - f(x), \quad \psi(x) = \frac{1}{2}\|x\|^2 - g(x)$$

where ψ is cvx and $T^{-1} = \nabla \psi$

Thm: W_p metrizes the weak* convergence,

$$\text{i.e. } \alpha_k \rightarrow \beta \iff W_p(\alpha_k, \beta) \rightarrow 0$$

Thm: $W_p(\alpha_k, \beta)$ is a proper distance in $\mathcal{P}(\mathcal{X})$

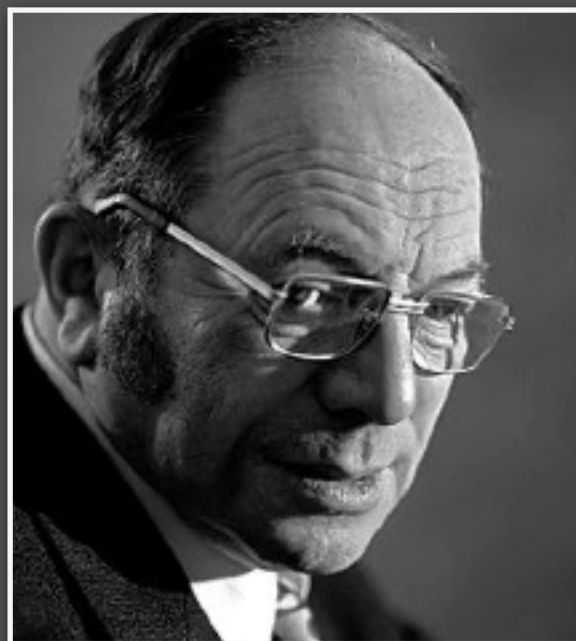
Wasserstein distance:

$$W_p(\alpha, \beta) = \left(\inf_{\pi \in \Pi(\alpha, \beta)} \int_X d(x, y)^p \, d\pi(x, y) \right)^{1/p}$$

$$W_1(\alpha, \beta) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\int f d\alpha - \int f d\beta \right)$$

$$W_2^2(\alpha, \beta) = \sup_{\phi \text{ convex}} \left(\int \phi(x) d\alpha(x) + \int \phi^* d\beta(y) \right)$$

Last Time



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$$\downarrow c(x, y) = d(x, y)^p$$

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Today

- A quick note on c -transforms, c -conjugates, and c -convexity
- **Deriving the dual** for the discrete Kantorovich problem
- Entropy Regularization and the **Sinkhorn algorithm**